

COMPUTING $\mathbb{S}\text{Bim}$ -MODULE CATEGORIES IN MAGMA

1. PROLOGUE

Outside of simple cases[†], decomposing Soergel bimodules by hand can be quite a difficult task. This difficulty remains present in decomposing the objects of other $\mathbb{S}\text{Bim}$ -module categories, such as the Lusztig–Vogan categories of Larson and Romanov ([LR22]). These notes will detail a general algorithm for decomposing such objects, which we were able to implement in Magma. Before that, however, we will briefly review Soergel bimodules and the Lusztig–Vogan categories. We will also see some examples of decompositions done by hand.

Typically, the category of Soergel bimodules is built from a *realization* $\mathcal{R} := (\mathfrak{h}, \{\alpha_s^\vee\}_{s \in S}, \{\alpha_s\}_{s \in S})$ of a Coxeter system (W, S) ; that is, a free $\mathbb{k}$ -module $\mathfrak{h}$ of finite rank, together with subsets $\{\alpha_s^\vee\}_{s \in S} \subset \mathfrak{h}$ and $\{\alpha_s\}_{s \in S} \subset \mathfrak{h}^* := \text{Hom}(\mathfrak{h}, \mathbb{k})$ for which

- (i). $\alpha_s(\alpha_s^\vee) = 2$;
- (ii). the map given by $\rho(s) : v \mapsto v - \alpha_s(v)\alpha_s^\vee$ for all $v \in \mathfrak{h}$ extends to a representation $\rho : W \curvearrowright \mathfrak{h}$;
- (iii). certain minor technical conditions are satisfied ([EMTW20, §5.7]).

Note that ρ induces a contragredient representation given by $\rho^*(s) : \gamma \mapsto \gamma - \gamma(\alpha_s^\vee)\alpha_s$ for all $\gamma \in \mathfrak{h}^*$.

Suppose we let $\mathfrak{h} := \bigoplus_{s \in S} \mathbb{R}\alpha_s^\vee$ and define elements $\{\alpha_s\} \subset \mathfrak{h}^* = \text{span}_{\mathbb{R}}\{\alpha_s : s \in S\}$ by

$$\alpha_s(\alpha_t^\vee) := \begin{cases} -2 \cos\left(\frac{\pi}{m_{st}}\right), & m_{st} \neq \infty; \\ -2, & m_{st} = \infty. \end{cases} \quad (1)$$

This defines what is known as the *geometric realization* $\mathcal{R}_{\text{geom}}$, and is what we will henceforth use. The *geometric representation* $W \curvearrowright \mathfrak{h}^*$ is always faithful, and it is positive-definite if and only if W is finite. However, it is not always reflection faithful, such as in the case of affine Weyl groups.

Remark 1. At first blush, this presents a problem, as Soergel’s theory (Soergel’s categorification theorem, Soergel’s conjecture, etc.) is only guaranteed to hold for *reflection faithful* realizations over an infinite field of characteristic not equal to 2. However, suppose we consider the *Kac–Moody realization*, given by taking $\mathfrak{h}$ to be the real vector space of minimal dimension with subsets $\{\alpha_s^\vee\} \subset \mathfrak{h}$ and $\{\alpha_s\} \subset \mathfrak{h}^*$ satisfying Equation (1). Soergel proved that this realization is always reflection faithful, and moreover Libedinsky showed that Soergel’s theory holds for the geometric representation if and only if it holds for this Kac–Moody representation ([EMTW20, §5.6]). In other words, the geometric representation will be sufficient for our purposes.

[†]...such as types A_1 and A_2 , which you can find my notes on here.

Remark 2. In [LR22], the *root datum realization* is used, which takes $\mathfrak{h}$ to be the Lie algebra of a maximal torus in a Borel subgroup of a connected complex reductive algebraic group (that is, the $\mathbb{k}$ -span of the cocharacter lattice of said maximal torus). This makes sense only for finite crystallographic Coxeter systems (finite Weyl groups). It is reflection faithful, producing an equivalent (additive monoidal $\mathbb{k}$ -linear) category by our last remark. Thus we will not mention it in the sequel.

Let $R := \text{Sym}(\mathfrak{h}^*)$, where $\deg(\mathfrak{h}^*) = 2$. Because $\{\alpha_s\}_{s \in S}$ forms an $\mathbb{R}$ -basis for $\mathfrak{h}^*$ in $\mathcal{R}_{\text{geom}}$, we have that $R = \mathbb{R}[\alpha_s : s \in S]$, with $\deg(\alpha_s) = 2$ for all $s \in S$. Define the fixed point subrings

$$R^I := \{f \in R : w(f) = f \text{ for all } w \in \langle I \rangle\}$$

for $I \subseteq W$ and consider the (R, R) -bimodules of the form

$$B_s := R \otimes_{R^s} R(1)$$

for $s \in S$, where (1) denotes a grading shift by 1. Given an expression $\underline{w} = (s_1, \dots, s_k)$ for $w \in W$, define the *Bott–Samelson bimodule* $BS(\underline{w})$ by

$$BS(\underline{w}) := B_{s_1} \otimes_R \cdots \otimes_R B_{s_k},$$

where we assume that $BS(\emptyset) = R$. Then we have the following definition.

Definition 3. (Soergel Bimodules). *The category of Soergel bimodules $\mathcal{SBim}(\mathcal{R})$ is defined to be the graded closure of the Karoubian envelope of the strictly full additive subcategory of $R\text{-GrMod-}R$ that is generated by Bott–Samelson bimodules.*

Put more simply, Soergel bimodules are direct summands of finite direct sums of grading shifts of Bott–Samelson bimodules. The morphisms are graded (R, R) -bimodule homomorphisms.

Consider now a finite index reflection subgroup $W_K \subseteq W$ (that is, a subgroup generated by elements $ws w^{-1}$, for $s \in S$ and $w \in W$) and let ${}^K W$ choose right coset representatives for $W_K \backslash W$. For each $w \in W$, define the *w-standard* (R^{W_K}, R) -bimodule R_w to be R as a vector space with the left action given by left multiplication and the right action given by $p \cdot_w f := pw(f)$, for all $p \in R_w$ and $f \in R$.

Definition 4. (Lusztig–Vogan Category). *The Lusztig–Vogan category $\mathcal{N}_{LV}^0(\mathcal{R}, K)$ is defined as*

$$\mathcal{N}_{LV}^0(\mathcal{R}, K) := \langle R_w \otimes_R X : w \in {}^K W, X \in \text{Ob}(\mathcal{SBim}(\mathcal{R})) \rangle_{\oplus, \ominus, (1)}.$$

That is, it is the additive, graded closure of the Karoubian envelope of the category generated by tensoring w-standard bimodules on the right with Soergel bimodules.

Remark 5. This definition arises when the two maximal tori chosen during the construction from [LR22] are of equal rank. It allows us to build $\mathcal{N}_{LV}^0$ for arbitrary Coxeter systems!

2. EXAMPLES

Now that we have been introduced to the protagonists of this story, let's see some examples. I claimed that these decompositions are difficult by hand, and I'd like to try to convince you of that.

Probably the limit of what I can do by hand is type A_2 , where $(W, S) = (S_3, \{s, t\})$. This is troublesome even in the case of Soergel bimodules! For example, in $\mathbb{S}\text{Bim}(A_2)$, we have

$$B_s \otimes_R B_t \otimes_R B_s \cong B_{sts} \oplus B_s.$$

This decomposition is given by inclusions $\iota_s : B_s \rightarrow B_s \otimes_R B_t \otimes_R B_s$ and $\iota_{sts} : B_{sts} \rightarrow B_s \otimes_R B_t \otimes_R B_s$ together with projections $\pi_s : B_s \otimes_R B_t \otimes_R B_s \rightarrow B_s$ and $\pi_{sts} : B_s \otimes_R B_t \otimes_R B_s \rightarrow B_{sts}$. In this case,

$$\iota_s : 1 \otimes_{R^s} 1 \mapsto \frac{1}{2}(1 \otimes_{R^s} (\alpha_t \otimes_{R^t} 1 + 1 \otimes_{R^t} \alpha_t) \otimes_{R^s} 1)$$

and

$$\iota_{sts} : 1 \otimes_{R^{s,t}} 1 \mapsto 1 \otimes_{R^s} 1 \otimes_{R^t} 1 \otimes_{R^s} 1$$

define the inclusions, while

$$\pi_s : 1 \otimes_{R^s} f \otimes_{R^t} g \otimes_{R^s} 1 \mapsto -2\partial_s(fg) \otimes_{R^s} 1$$

and

$$\begin{aligned} \pi_{sts} : 1 \otimes_{R^s} f \otimes_{R^t} g \otimes_{R^s} 1 \mapsto & P_s(f) \otimes_{R^{s,t}} P_s(g) + \\ & \partial_s(f) \frac{2}{3} \left(\frac{1}{2} \alpha_s + \alpha_t \right) \otimes_{R^{s,t}} P_s(g) + \\ & \partial_s(f) \otimes_{R^{s,t}} \frac{4}{3} \left(\frac{1}{2} \alpha_s + \alpha_t \right) P_s(g) + \\ & P_s(f) \otimes_{R^{s,t}} \frac{2}{3} \left(\frac{1}{2} \alpha_s + \alpha_t \right) \partial_s(g) + \\ & P_s(f) \frac{4}{3} \left(\frac{1}{2} \alpha_s + \alpha_t \right) \otimes_{R^{s,t}} \partial_s(g) + \\ & \partial_s(f) \frac{1}{3} \left(\frac{1}{2} \alpha_s + \alpha_t \right) \otimes_{R^{s,t}} \frac{1}{3} \left(\frac{1}{2} \alpha_s + \alpha_t \right) \partial_s(g) + \\ & \partial_s(f) \otimes_{R^{s,t}} \frac{2}{3} \alpha_s^2 \partial_s(g) + \\ & \partial_s(f) \frac{2}{3} \alpha_s^2 \otimes_{R^{s,t}} \partial_s(g) - \\ & \partial_s(f) \frac{4}{9} (\alpha_s^2 + \alpha_s \alpha_t + \alpha_t^2) \otimes_{R^{s,t}} \partial_s(g) \end{aligned}$$

define the projections. It's perhaps not too hard to see how this could get difficult quickly in larger examples! This task remains difficult for our Lusztig–Vogan categories; in fact, since $\mathcal{N}_{LV}^0(\text{SU}(2, 1))$ consists of (R^s, R) -bimodules, truncating the inclusions above by cutting off the B_s at the front will induce the decomposition $R \otimes_R B_t \otimes_R B_s \cong (R \otimes_R B_{sts}) \oplus R$ as (R^s, R) -bimodules. Meanwhile, the projection of $R \otimes_R B_t \otimes_R B_s$ into R is given by

$$f \otimes_{R^t} g \otimes_{R^s} h \mapsto \partial_s(fg)h,$$

while the projection into $R \otimes_R B_{sts}$ is some (computable) mess I haven't written down. Of course, this only works for $R \otimes_R B_t \otimes_R B_s$; for $R_t \otimes_R B_s \otimes_R B_t \cong (R \otimes_R B_{sts}) \oplus R_t$, we are not so fortunate, and end up with (for example) the scarier projection into R_{ts} of the form

$$f \otimes_{R^t} g \otimes_{R^s} h \otimes_{R^t} k \mapsto ts(\delta_t(st(f)s(g)h)k).$$

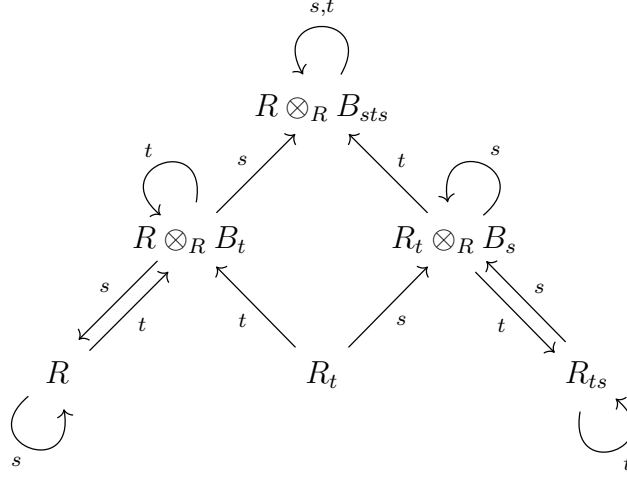


FIGURE 1. The W -graph for $\mathcal{N}_{LV}^0(\mathrm{SU}(2, 1))$, showing its module structure.

Doing these examples by hand relies both on making educated guesses for the inclusions and a trick for the projections that only works for modules with two indecomposable summands. While we do have a few tools for determining indecomposability without explicitly computing the idempotents, even these unfortunately do not get us very far in more complicated settings!

3. THE ALGORITHM

Suppose $R^{W_K} = \mathbb{R}[x_1, \dots, x_m] \subseteq R$, graded in degree 2, and consider some graded Krull–Schmidt (R^{W_K}, R) -bimodule Q that is graded free of finite rank as a right R -module with basis $\{b_1, \dots, b_n\}$. All Soergel bimodules and objects in our Lusztig–Vogan categories satisfy this. We want to determine the indecomposable summands of Q ; this is equivalent to determining its primitive idempotents.

Let’s ignore the grading for a moment. As a free right R -module, we know that $Q \cong R^{\oplus n}$ and hence that $\mathrm{End}_{\mathrm{Mod}\text{-}R}(Q) \cong M_n(R)$. As a result, we can encode the left action of R^{W_K} via the basis $\{A_k\}_{k=1}^m$ of $n \times n$ matrices satisfying $A_k f = x_k \cdot f$ for all $f = b_1 \cdot f_1 + \dots + b_n \cdot f_n \in Q$. More explicitly, we may write $A_k := (a_{ij}^k)_{i,j=1}^n$, where a_{ij}^k is the coefficient of b_i in $x_k \cdot b_j$. Then $\mathrm{End}_{R^{W_K}\text{-Mod}\text{-}R}(Q)$ is precisely the subset of $M_n(R)$ consisting of matrices that commute with every A_k . Finally, in order to reintroduce the grading, we need only enforce homogeneity on each matrix M ; that is, we require some $d \in \mathbb{N}$ such that every Mb_k is homogeneous in degree d . The set of degree d endomorphisms is

$$\mathrm{End}^d(Q) \cong \left\{ M = (m_{ij})_{i,j=1}^n \in M_n(R) : \begin{array}{l} MA_k - A_k M, \forall k \in \{1, \dots, m\}; \\ \deg(m_{ij}) = d - \deg(b_i) + \deg(b_j), \forall m_{ij} \neq 0 \end{array} \right\}.$$

Thus we are interested in the set

$$\mathrm{Idem}(Q) := \{E \in \mathrm{End}^0(Q) : E^2 = E\},$$

where the condition $E^2 = E$ enforces $d = 0$. These conditions give us a system of equations that we can solve to find the idempotents for Q . The primitive idempotents are then the orthogonal idempotents of minimal rank whose ranks sum to the rank of Q .

Observe what we have just done: if all one cares about are its idempotents, Q may be encoded as a basis $\{b_1, \dots, b_n\}$ together with a set of left action matrices $\{A_k\}_{k=1}^m$. This lays the foundation for our algorithm. In practice, because this basis is built into the action matrices, we really only need to keep track of the grading of each basis vector.

Suppose we now have an arbitrary (R, R) -bimodule Q that is free of finite rank as an R -module with basis $\{b_1, \dots, b_n\}$ and left action matrices $A_k : f \mapsto x_k \cdot f$. One can show that, using the basis

$$b_1^s := 1 \otimes_{R^s} 1, \quad b_2^s := \frac{1}{2}(\alpha_s \otimes_{R^s} 1 + 1 \otimes_{R^s} \alpha_s),$$

for B_s , the bimodule $Q \otimes_R B_s$ admits a basis $\{b_i \otimes_R b_1^s\}_{i=1}^n \sqcup \{b_i \otimes_R b_2^s\}_{i=1}^n$ and left action matrices

$$A_k^s := \begin{pmatrix} s(A_k) & 0 \\ \partial_s(A_k) & A_k \end{pmatrix}$$

for all $k \in \{1, \dots, m\}$. Thus we have also a way of encoding the product $Q \otimes_R B_s$.

With all of this in mind, we are ready to understand how our algorithm works. We begin by encoding the w -standard bimodule R_w for each $w \in {}^K W$. These are all rank 1 and are hence encoded by the degree 0 basis vector $1 \in R$ together with the left action scalars $\{w(x_k)\}_{k=1}^m$. From here, we proceed “layer-by-layer”, tensoring by B_s for each $s \in S$. After tensoring with some B_s , we find the primitive idempotents by solving the aforementioned system of equations. Each idempotent matrix admits some basis $\{b_1, \dots, b_r\}$ for its image (where the standard basis vectors are taken to be the basis for the module being decomposed), along with a matrix P such that PI is the $r \times r$ identity matrix (where I is the matrix whose columns are the basis vectors $\{b_1, \dots, b_r\}$). Then our new indecomposable is encoded by $\{b_1, \dots, b_r\}$ together with $\{PA_k^s I\}_{k=1}^m$. We finally check if this indecomposable is isomorphic to one we’ve seen before and then repeat the process.

REFERENCES

- [EMTW20] Elias, B., Makisumi, S., Thiel, U., and Williamson, G., *Introduction to Soergel Bimodules*, vol. 5, RSME Springer Series, Springer Nature Switzerland, 2020.
- [LR22] Larson, S. and Romanov, A., *A categorification of the Lusztig–Vogan module* (2022), DOI: 10.48550/arXiv.2203.09007.